

Episode organization

Split source episodes before windowing

1. Validate source replay

Simulation log

actions $a_{0:T}$
states $s_{0:T}$
...

Replay record

actions $a_{0:T}$
states $s_{0:T}$
...

Check source
+ replay hash
(same task, same reset)

Unique
episode

2. Split episode identities

Split by episode identity e

Train

(40 episodes)

Development

(10 held-out episodes)

3. Build chunks within each split

Sliding window (train)

...

...

$H = 16 \cdot \text{stride} = 8$

Sliding window (dev)

...

...

$H = 16 \cdot \text{stride} = 8$

490 dev windows
(unaugmented)

1960 train windows; source episodes stay separate.

Record contract

Current observation and executed action

Current input o_t (first three fields enter the policy)

	Robot state	q_t	14 dimensions
	Object position	p_t	3 / 6 dimensions
	Elapsed time	t/T	1 dimension
	Source episode	e	audit metadata
	Branch endpoint	u	audit metadata
	Sampling seed	seed	audit metadata



Executed action

Y_t

16×14

Source and replay metadata

Recovery rollout

Pulse + success check

seed
branch seed



$(\tilde{o}_t, Y_t)$

Reset

s_0

Figs. 3–4

Traceable windows

One accepted replay, up to four windows

≤ 4 complete windows after the pulse

Replay $\tilde{o}_t$

Replay $\tilde{o}_t$



Y_t

$t = u$



source $e \cdot$ endpoint $u \cdot$ start u

Replay $\tilde{o}_t$



Y_t

$u + 8$



source $e \cdot$ endpoint $u \cdot$ start $u+8$

Replay $\tilde{o}_t$



Y_t

$u + 16$



source $e \cdot$ endpoint $u \cdot$ start $u+16$

Replay $\tilde{o}_t$



Y_t

$u + 24$



source $e \cdot$ endpoint $u \cdot$ start $u+24$

$(\tilde{o}_t, Y_t)$
 $t \in U(u)$